MESSOU Franck

AI / Machine Learning Engineer | LLMs & GenAI | NLP & Speech | MLOps & Model Serving

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Summary

Machine learning engineer and PhD researcher who ships applied AI end-to-end: on-device fine-tuned small language models with cloud LLM-API fallback (Gemini, GPT), speech-to-text and NLP pipelines (summarization, keyword extraction, translation across 20 languages), and IoT/edge sensor intelligence. Solid MLOps foundation: model fine-tuning (LoRA / PEFT), distributed training, and low-latency inference serving on dedicated multi-GPU infrastructure. First-author of 2 Q1 journal papers and 5 IEEE conference papers.

Technical Skills

Programming: Python, PyTorch, SQL, Bash

LLMs & GenAI: LLM / sLM fine-tuning (LoRA, PEFT), Hugging Face Transformers, RAG, prompt engineering, vLLM; Gemini, GPT / OpenAI, Claude, Llama, Qwen, Phi, SmolLM

NLP & Speech: Speech-to-text (ASR), summarization, keyword extraction, machine translation (multilingual), text classification

ML & Deep Learning: Deep learning, time-series forecasting, anomaly detection, graph neural networks, scikit-learn

MLOps & Serving: Model serving (vLLM, FastAPI), distributed training (DDP), CUDA, Docker, inference / latency optimization, model checkpointing, GCP

Data: pandas, NumPy, ETL data pipelines, feature engineering, PostgreSQL, MongoDB, Firestore

Experience

Machine Learning Engineer — Oct 2025 – Jul 2026

D'isum Inc. — Tokyo, Japan

- Built ETL pipelines turning raw sensor streams from 50+ engines and 200+ batteries into analytics-ready datasets for predictive models.
- Developed and deployed classical ML models for engine sound analysis and battery time-series health monitoring (anomaly and fault detection) in client systems.
- Built on-device iOS / Android sensor apps (Swift, Java) for real-time data capture and edge inference.

AI / ML Researcher (Doctoral) — Sep 2025 – Present

YuLab, Hosei University — Tokyo, Japan

- Built LLM / small-language-model fine-tuning pipelines (Spec2Llama, Spec2LLM, LLM4Imp) adapting frozen LLMs and compact sLM backbones (SmolLM, Phi, Qwen) for time-series forecasting and imputation on edge, published at IEEE ICCE / ICC 2026.
- Provision and operate the lab's multi-GPU server end-to-end: distributed training (DDP), model checkpointing, and vLLM inference serving in Docker, shared across the research team.
- Graph-neural-network models for power-grid anomaly detection (research; see Publications).

ML Researcher (Master's) — 2023 – Sep 2025

YuLab, Hosei University — Tokyo, Japan

- Built transformer time-series forecasting models (TAPformer, TSFormer, CNN-BiLSTM) for long-term electricity load forecasting in IoT systems, outperforming standard CNN / LSTM baselines (published in *Internet of Things*, IF 7.6, Q1).
- Won the IEEE VTS Tokyo / Japan Chapter 2024 Student Paper Award (VTC2024-Fall, Washington D.C.).

Generative AI Intern · TICAD9 Booth — Jul 2024 · Aug 2025

Japan AI Consulting (J-AIC) / JICA — Osaka · Yokohama

- Built GenAI prototypes for the consulting practice using Flask, PostgreSQL, and LLM APIs (Jul 2024 internship).
- Represented J-AIC and presented applied ML / GenAI prototypes at JICA's TICAD9 booth, Yokohama (Aug 2025).

Projects

AIReco – AI meeting assistant (launching on app stores) | on-device fine-tuned sLM, Gemini / GPT, ASR, NMT, GCP — 2025 – 2026

- Records and transcribes meetings, then generates summaries, keywords, and translations across 20 languages.
- Hybrid routing between an offline fine-tuned sLM and cloud LLM APIs for cost, latency, and privacy.

LLM time-series reprogramming (Spec2Llama / Spec2LLM / LLM4Imp) | LLMs, sLMs, PEFT, PyTorch 2024 – Present

- Adapt frozen language models for forecasting and imputation; published at IEEE ICCE 2026 and ICC 2026.

Multi-GPU MLOps stack | CUDA, DDP, vLLM, Docker, Linux 2023 – Present

- Reusable training and inference-serving infrastructure provisioned and operated end-to-end.

Public code: github.com/mesabo | github.com/hosei-university-iist-yulab

Selected Publications

- **MFJ Aboya** et al., "TAPformer: Long-Term Electricity Load Forecasting in IoT Systems," *Internet of Things*, Elsevier, 2026. (IF 7.6, Q1)
- **MFJ Aboya** et al., "Spec2Llama: Spectral-Aware Forecasting for Versatile Time Series," *Expert Systems with Applications*, Elsevier, 2026. (IF 7.5, Q1)
- **MFJ Aboya** et al., "Hybrid Attention-Based CNN-BiLSTM for Short-Term Load Forecasting in IoT," *IEEE VTC2024-Fall*, 2024. (Student Paper Award)
- **MFJ Aboya** et al., "TSFormer: Temporal-Aware Transformer for Multi-Horizon Forecasting," *IEEE ICUFN 2025*.
- **MFJ Aboya** et al., "Spec2LLM: Spectral-to-Language Reprogramming for Power Transformer Temperature Forecasting," *IEEE ICCE 2026*.
- **MFJ Aboya** et al., "LLM4Imp: Frozen LLMs with Spectral Prompts for Time-Series Imputation," *IEEE ICC 2026*.
- **MFJ Aboya** et al., "Large Language Models for Temporal Sensor Networks: Graph-Constrained Multi-LLM Ensembles," *IEEE VTC2026-Spring*.

Education

PhD, Applied Informatics Sep 2025 – Apr 2027 (early)
Hosei University Tokyo, Japan

MSc, Applied Informatics 2023 – Sep 2025
Hosei University Tokyo, Japan

Languages

Spoken: French (native), English (fluent), Japanese (simple daily conversation, ongoing study)